EBS performance is more complicated. Performance depends on the type of EC2 instance as well as the EBS volume type. EC2 instances with EBS optimization benefit by having dedicated bandwidth to their EBS volumes. Table 8.1 gives an overview of EC2 instance types that are EBS-optimized by default. Some older instance types can be optimized for an additional hourly charge whereas others do not support EBS optimization at all. Input/output operations per second (IOPS) are measured using a standard 16 KB I/O operation size.

Table 8.1 What performance can be expected from modern instance types? Your mileage may vary.

Use case	Instance type	Baseline bandwidth (Mbps)	Maximum bandwidth (Mbps)
General purpose	m6a.large–m6a.48xlarge	531–40,000	6,666–40,000
Compute optimized	c6g.medium–c6g.16xlarge	315– 9,000	4,750–19,000
Memory optimized	r6i.large–r6i.32xlarge	650–40,000	10,000–40,000
Memory and network optimized	x2idn.16xlarge–x2idn.32xlarge	40,000–80,000	40,000–80,000

WARNING Performance depends heavily on your workload: read versus write, as well as the size of your I/O operations (a bigger operation size equates to more throughput).

Depending on your storage workload, you must choose an EC2 instance that can deliver the bandwidth you require. Additionally, your EBS volume must be balanced with the amount of bandwidth. Table 8.2 shows the different EBS volume types and how they perform.